LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	23
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Aircraft performance: a/c characteristics, engine Characteristics, minimum speed, maximum rate of climb, minimum Angle of climb, range and E
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Aircraft performance: a/c characteristics, engine Characteristics, minimum speed, maximum rate of climb, minimum Angle of climb, range and Endurance.
5. Specific Objectives	6.1.3.1 Trainee will be able to describe flight performance of aircraft.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of theoretical performance curves; Instructor supervision of diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary factors affecting aircraft performance?
(b) Define the relationship between lift and weight during steady flight.
(c) How does engine power output vary with altitude?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Aircraft and Engine Characteristics • Minimum Speed and Stall • Climb Performance Metrics	L & Ds	PPT, IB	15 min
(b)	• Maximum Rate of Climb Analysis • Minimum Angle of Climb Analysis • Range and Endurance Calculations	L & Ds	PPT, PDF	20 min
(c)	• Performance Trade-offs in Flight • Impact of Weight and Density Altitude • Operational Implications of Performance Limits	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between maximum rate of climb and minimum angle of climb.
(b) Describe how aircraft weight affects the minimum speed required for flight.
(c) Why is endurance distinct from range in terms of fuel consumption?

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of range and endurance discussed in this lesson will be applied during the take-off and landing procedures covered in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Take-off and landing. Definition factors effecting take off distances. General performance, landing distance. Aircraft landing circuits.	Refer to the master syllabus reference list	LP 24

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min